Integrated Postal Architecture
Trazar Mail Tracking

Precision Postal Software Design

The Mail Management modules of the Integrated Postal Architecture (IPA) are known by the product name of *Trazar* and are designated by the use of the *Trazar* logo.

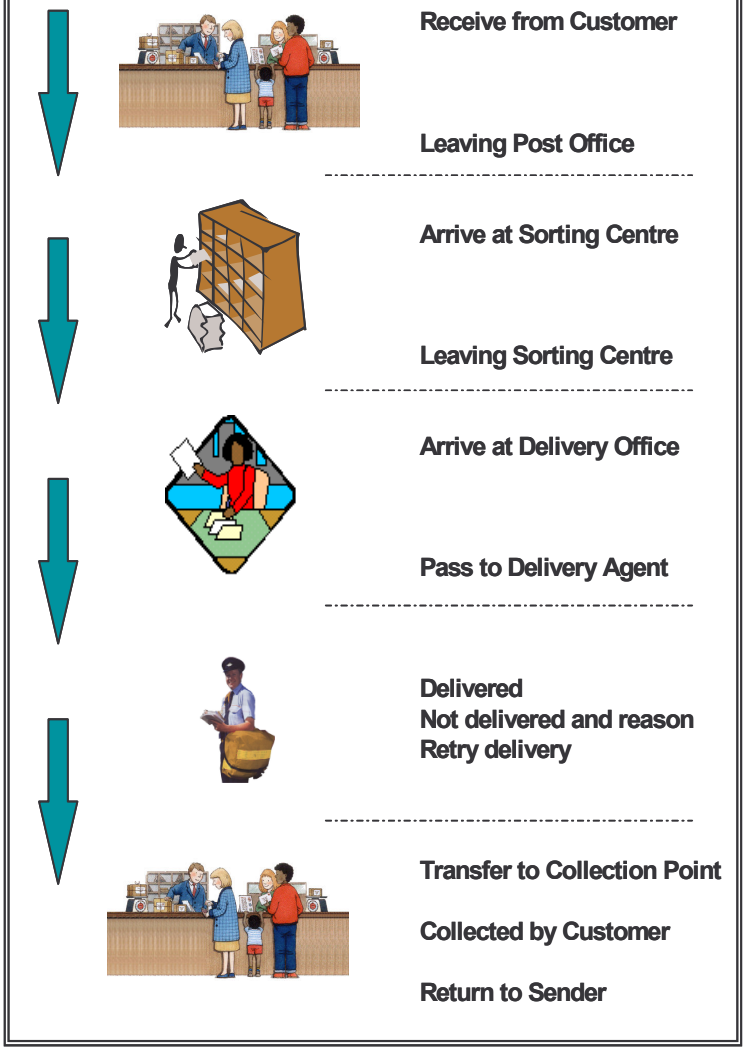
Trazar encompasses four modules, the proposal of which will depend upon which facilities required.

TRAZAR also integrates to the UPU's International Postal System to provide a single seamless system that manages and reports on all mail movements.

The *TRAZAR* modules enable a Postal Enterprise to track premium mail items from end-to-end, providing both customers and management with valuable information relating to the location of items and the quality of the service being provided.

The *TRAZAR* module includes:

- ⇒ End-to-end premium mail item tracking
- ⇒ "Receive from Customer" event capture embedded in *CORVUS* counters system
- ⇒ Receptacle tracking and management
- ⇒ Despatch tracking and management
- ⇒ Choice of captured lifecycle item events
- ⇒ Automatic production of receptacle manifests and despatch manifests
- ⇒ Interface to UPU's International Postal System (IPS)
- ⇒ Delivery management
- ⇒ Quality-of-service measurement reports
- ⇒ Optional use of 2D barcodes offers less reliance on the network



Lifecycle item events

The *TRAZAR* system enables the capture of a variable range of movement events in the lifecycle of each mail item, receptacle and despatch. Within the post office, the "Receive from Customer" will be captured at the counter or back office using *CORVUS* or *TRAZAR*. At every point of movement of the item, its ID is scanned and recorded on the central tracking database.

The *TRAZAR* system allows the Post to select which "Events" are captured for each mail product. For example, a registered mail service for which the Post charges a small financial uplift over the normal mail price may only warrant the capture of the "Receive from Customer" event and the "Delivered" events, such that the Post can provide a proof of receipt from customer and a proof of delivery. On the other hand, a high value service such as EMS may require more accurate tracking by the capture of more lifecycle events. The events captured for each service will be agreed with the Post for each automated service.

The example shown here is a common set of events captured for a high premium mail service.

TRAZAR allows the events captured to be configured for each different mail service being managed by the system

IPA = Cost-effective, Flexible, Integrated Solutions for the Postal Industry

Event Capture

Events are captured at every point where an entity changes location or “ownership”. Following receipt of an item at the counter, the information relating to the item is sent to the central tracking database.

In the back office receptacles (mailbags) are created and filled with the relevant mail items. A bag label is printed and a Receptacle Manifest listing all items in the bag.

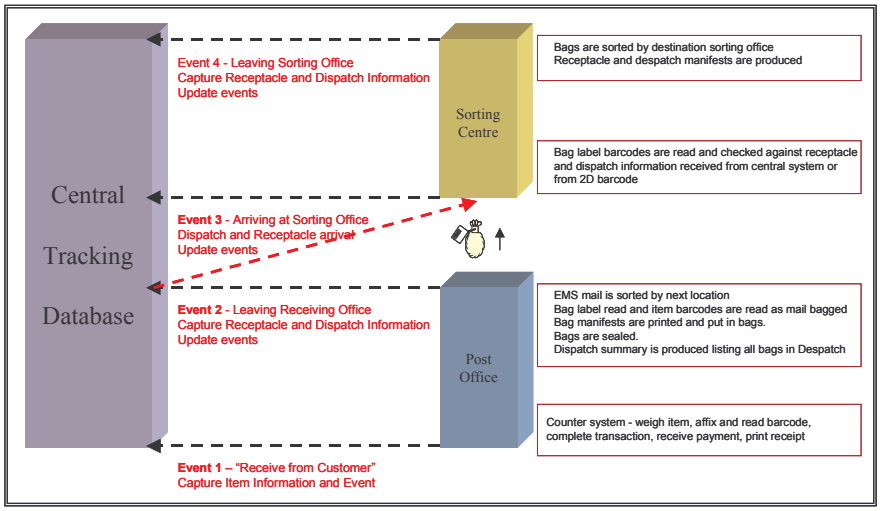
Multiple bags sent between locations are defined as despatches, and a Despatch Summary is printed showing a list of receptacles.

When a despatch leaves a location the central tracking database is updated with the relevant events relating to the item, the receptacle and the despatch.

If a reliable communications network is in place to link the automated locations, an electronic update will take place at the next destination of the despatch. This enables the despatch to be completely verified upon arrival at its destination.

If, however, there is no network, or a network breakdown has occurred, the details of the despatch, the receptacle and all mail items can be verified by reading the 2D barcode on the Despatch Summary and the Receptacle Manifests. The use of 2D barcodes in this way enables Posts to commence the implementation of an automated mails network without the necessity for a countrywide network to be in place.

The diagram shows the basic principal of updating the central database and pre-notifying the next locations.



Item Identification

All entities to be tracked within a mail network require an electronically readable ID. This includes mail items, receptacles and despatches, and is normally a bar-coded ID number to an accepted standard.

For international items all IDs should adhere to the UPU defined standards in order to allow for international tracking and management of the items. Item identification normally uses a simple code 128 or code 39 bar-coded label. It is essential that the ID is unique.

In order to assist Posts in the provision of label stock, PSL has provided a system to the UPU for the generation of unique label stock for premium mail items.

Sample labels are shown here:



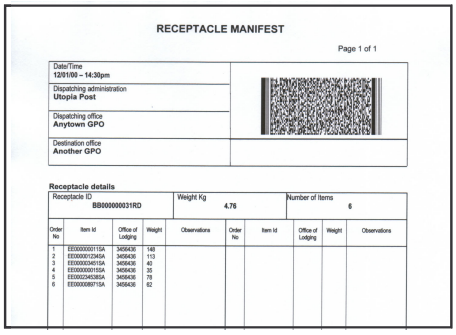


PSLs barcode label generation software can be used by a Post to generate unique bar-coded ID labels for all of its international and domestic mail products, thus minimising the costs involved in providing trackable mail items.

2D barcoded documents

TRAZAR allows the use of 2D barcode identifiers on Receptacle Manifests and Despatch Summaries. This facility reduces the reliance on the communications network to provide pre-notification of despatch arrivals.

The 2D barcode encodes the full details of the manifest such that the contents of a receptacle can be captured with a single scan.



The Right Database

As with other modules of the Integrated Postal Architecture, *TRAZAR* is database independent.

Being ODBC compliant, *TRAZAR* will operate on a range of database management systems from MS Access, through MS SQL Server, Oracle and for the largest users *TRAZAR* offers a number of Data Warehousing solutions including Teradata.

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